Introduction to Causal Models

Moo K. Chung
University of Wisconsin-Madison
mkchung@wisc.edu

1 Introduction

Causality is a fundamental concept in science, aiming to determine whether one event influences another. In time series analysis, **Granger causality**, introduced by Clive Granger in 1969, provides a statistical framework for assessing whether past values of one time series help predict another (Granger, 1969).

Unlike traditional notions of causality in physics or philosophy, Granger causality is based on predictability: if including past values of one time series improves the forecast of another, it is said to have a causal influence. For this reason, it is often referred to as **predictive causality**. Similarly, SEM captures **associational causality** through contemporaneous linear relationships. Both differ from **interventional causality** in structural causal models, which is defined through external interventions and counterfactual reasoning.

Granger causality has wide applications in economics, neuroscience, climate science, and engineering (Barnett and Seth, 2014). In a multivariate time series, where multiple variables evolve over time, it extends naturally to examine interactions among multiple signals and is one of the most commonly used causal models in brain network analysis. Granger causality is typically implemented using Vector Autoregressive (VAR) models.

2 Vector Autoregressive (VAR) Model

The **Vector Autoregressive (VAR)** model is a fundamental framework for modeling multivariate time series, where multiple variables evolve over time and potentially influence each other (Gorrostieta et al., 2012; Melnyk and Banerjee, 2016; Primiceri, 2005). Unlike univariate autoregressive models that describe the evolution of a single time series, the VAR model extends this concept to a system of interdependent time series. This makes it particularly useful for capturing dynamic relationships in fields such as economics, neuroscience, and climate science.

Consider a system of N time series denoted as

$$X(t) = [X_1(t), X_2(t), \ldots, X_N(t)]^\top,$$

where $X_i(t)$ represents the time series of the i-th region at time t. The VAR model of order P, denoted as VAR(P), expresses each time series as a linear combination of its own past values and the past values of all other variables in the system. $X(t) \in \mathbb{R}^N$ is a time-varying vector. This model is formulated as

$$X(t) = \sum_{p=1}^{P} A_p X(t-p) + \varepsilon(t), \tag{1}$$

where A_p are $N \times N$ coefficient matrices that capture the influence of past values at lag p, and $\varepsilon(t)$ is an N-dimensional vector of error terms, assumed to be normally distributed with zero mean and a covariance matrix Σ, i.e.,

$$\varepsilon(t) \sim \mathcal{N}(0, \Sigma).$$

The number of lags P is a crucial hyperparameter that determines how far back in time the model considers past information. Expanding each component, we can write

$$X_i(t) = \sum_{p=1}^{P} \sum_{j=1}^{N} A_{ij}^{(p)} X_j(t-p) + \varepsilon_i(t), \tag{2}$$

where $A_{ij}^{(p)}$ represents the effect of the j-th variable at lag p on the i-th variable. Note $A_p = (A_{ij}^{(p)})$. If $A_{ij}^{(p)}$ is significantly different from zero, this indicates a directional relationship where past values of $X_j(t)$ contribute to predicting $X_i(t)$, which is the foundation for testing Granger causality.

The estimation of VAR model parameters is typically performed using least squares estimation:

$$\min_{A_1, \cdots, A_P} \sum_{t=P+1}^{T} \|X(t) - \sum_{p=1}^{P} A_p X(t-p)\|^2. \tag{3}$$

Instead of estimating the coefficients iteratively, which will compound errors, we can estimate the entire coefficient matrix simultaneously in a single step by reformulating the problem in matrix form. Let

$$A = [A_1, A_2, \ldots, A_P] \in \mathbb{R}^{N \times (NP)}, \tag{4}$$

which stacks all lag-specific coefficient matrices horizontally. Next, define the regressor matrix:

$$X_{\text{reg}}(t) = [X(t-1)^\top, X(t-2)^\top, \ldots, X(t-P)^\top]^\top \in \mathbb{R}^{(NP) \times 1}. \tag{5}$$

This allows the VAR model to be rewritten in a compact form as

$$X(t) = A X_{\text{reg}}(t) + \varepsilon(t), \quad t = P+1, P+2, \cdots T \tag{6}$$

where $\varepsilon(t)$ is the residual error term assumed to follow a multivariate normal distribution with zero mean and covariance Σ, i.e.,

$$\varepsilon(t) \sim \mathcal{N}(0, \Sigma). \tag{7}$$

Stacking the equations over all time points from $P+1$ to T, we define

$$Y = [X(P+1), X(P+2), \ldots, X(T)] \in \mathbb{R}^{N \times (T-P)}, \tag{8}$$

where Y represents the matrix of observed time series from $P+1$ to T. Similarly we define

$$\mathbf{X}_{\text{reg}} = [X_{\text{reg}}(P+1), X_{\text{reg}}(P+2), \ldots, X_{\text{reg}}(T)] \in \mathbb{R}^{(NP) \times (T-P)}, \tag{9}$$

Then we can write the VAR as a single matrix equation as

$$Y = A\mathbf{X}_{\text{reg}} + E, \tag{10}$$

where E is the residual error matrix

$$E = [\varepsilon(P+1), \varepsilon(P+2), \ldots, \varepsilon(T)] \in \mathbb{R}^{N \times (T-P)}. \tag{11}$$

To estimate A, we minimize the sum of squared residuals:

$$\widehat{A} = \arg\min_{A} \|Y - A\mathbf{X}_{\text{reg}}\|_F^2, \tag{12}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The least squares solution is given by

$$\widehat{A} = Y\mathbf{X}_{\text{reg}}^{\top} \left(\mathbf{X}_{\text{reg}}\mathbf{X}_{\text{reg}}^{\top}\right)^{-1}. \tag{13}$$

Once the coefficient matrix $\widehat{A}$ is obtained, the predicted values for all time points can be computed as

$$X_{\text{pred}} = \widehat{A}\mathbf{X}_{\text{reg}}. \tag{14}$$

This formulation ensures that all parameters are estimated simultaneously in a single step, minimizing the total squared error over all time points. The simultaneous estimation method provides a more stable and efficient approach compared to iterative forecasting methods, particularly when dealing with high-dimensional time series data.

The Vector Autoregressive (VAR) model requires past values of the time series $X(1), \ldots, X(P)$ to make predictions. The selection of the optimal lag order P is an essential step in model fitting. Choosing too small a value may result in missing important dependencies, whereas an excessively large P may introduce overfitting. The lag order is typically determined using information criteria such as the Akaike Information Criterion (AIC) or the Bayesian Information Criterion (BIC), which balance model complexity with goodness of fit.

Despite its flexibility and interpretability, the VAR model assumes stationarity and its statistical properties of the time series remain constant over time.

Additionally, VAR models assume linear relationships, which may not fully capture nonlinear dependencies present in brain network data.

The VAR framework provides a powerful foundation for analyzing multivariate time series and forms the basis for testing Granger causality. By capturing interactions among multiple variables, it serves as an essential tool for investigating directional dependencies and understanding the underlying structure of complex dynamic systems.

3 Granger Causality

Granger causality provides a statistical framework for determining whether the past values of one time series improve the prediction of another within a multivariate time series setting. Unlike traditional notions of causality, which imply a mechanistic link, Granger causality is based purely on predictive ability: if including past values of a variable $X_j(t)$ significantly reduces the prediction error of another variable $X_i(t)$, then $X_j(t)$ is said to **Granger-cause** $X_i(t)$.

Given a multivariate system modeled by a vector autoregressive process of order P (VAR(P)):

$$X_i(t) = \sum_{p=1}^{P} \sum_{j=1}^{N} A_{ij}^{(p)} X_j(t-p) + \varepsilon_i(t), \tag{15}$$

where $A_{ij}^{(p)}$ are the autoregressive coefficients capturing the influence of variable j on variable i at lag p, and $\varepsilon_i(t)$ is a zero-mean white noise process with covariance Σ. To test whether $X_j(t)$ Granger-causes $X_i(t)$, we compare two models:

- **Full model:** The standard VAR(P) equation, where $X_i(t)$ depends on the past values of all variables, including $X_j(t-p)$, given by:

$$X_i(t) = \sum_{p=1}^{P} \sum_{j=1}^{N} A_{ij}^{(p)} X_j(t-p) + \varepsilon_i(t). \tag{16}$$

- **Restricted model:** A modified VAR(P) equation where the past values of $X_j(t)$ are excluded from the predictors:

$$X_i(t) = \sum_{p=1}^{P} \sum_{l \neq j} A_{il}^{(p)} X_l(t-p) + \varepsilon_i'(t). \tag{17}$$

  We then determine whether the exclusion of $X_j(t-p)$ significantly worsens the prediction of $X_i(t)$. If the residual error in the restricted model increases significantly compared to the full model, we conclude that $X_j(t)$ provides useful predictive information for $X_i(t)$ and therefore Granger-causes it. To statistically test this, we define the hypotheses as follows:

$$H_0 : A_{ij}^{(p)} = 0, \quad \text{for all } p \in \{1, \ldots, P\}, \tag{18}$$

$$H_A : A_{ij}^{(p)} \neq 0, \quad \text{for at least one } p. \tag{19}$$

Under the null hypothesis H_0, the past values of $X_j(t)$ do not contribute to predicting $X_i(t)$, implying no Granger causality. If the null hypothesis is rejected, it suggests that past values of $X_j(t)$ improve the prediction of $X_i(t)$, supporting the presence of Granger causality. Thus, we use the F-statistic computed as

$$F = \frac{\big(\mathrm{RSS}_{\text{restricted}} - \mathrm{RSS}_{\text{full}}\big)/P}{\mathrm{RSS}_{\text{full}}/\big(T - P - NP\big)} \sim F_{P,\,T-P-NP}, \tag{20}$$

where: $\mathrm{RSS}_{\text{restricted}}$ is the residual sum of squares for the restricted model, $\mathrm{RSS}_{\text{full}}$ is the residual sum of squares for the full model, and NP is the number of regressors in the full model for one equation.

Under H_0, the corresponding p-value is computed as

$$p = 1 - F_{P,\,T-P-NP}(F), \tag{21}$$

with the cumulative distribution function $F_{P,\,T-P-NP}$.

3.1 Granger Causality Matrix

For large numbers of regions (N), the test results are often summarized in a *Granger causality matrix* $G = (G_{ij})$, where each entry G_{ij} is the p-value of the test for whether the past values of $X_j(t)$ significantly improve the prediction of $X_i(t)$. Smaller p-values indicate stronger evidence of Granger causality. Thus, we can construct a weighted directed graph $W = (w_{ij})$ by defining

$$w_{ij} = 1 - G_{ij},$$

so that larger values of w_{ij} correspond to stronger causal influence from X_j to X_i. In this representation, the edge weights directly reflect the strength of the detected Granger causal relationships. This provides a convenient way to visualize directional relationships among brain regions. In neuroscience, Granger causality is widely used to infer functional connectivity from fMRI and EEG data, helping to reveal how activity propagates across brain networks. Beyond neuroscience, it is also a standard tool in econometrics, where it is applied to examine predictive relations among financial indicators such as interest rates, exchange rates, and stock prices.

4 Structural Equation Model

While Granger causality provides a predictive notion of temporal influence by testing whether past values of one variable improve the forecast of another, an alternative and complementary approach is provided by the *Structural Equation Model (SEM)* framework (Bollen, 1989; Rijsdijk and Sham, 2002; Schmitt et al., 2009). SEM is rooted in econometrics and psychometrics, and has been widely adopted in neuroscience for analyzing brain networks from fMRI and EEG data. Unlike Granger causality, which is inherently temporal and based on lagged dependencies, SEM focuses on *contemporaneous relationships* among variables at the same time point. In SEM, directed edges represent direct causal effects among observed variables, and the model is built as a system of simultaneous linear equations.

4.1 Structural Equation Model

Consider a system of N observed variables at time t, denoted by

$$X(t) = [X_1(t), X_2(t), \ldots, X_N(t)]^\top \in \mathbb{R}^N.$$

The SEM formulation assumes that each variable $X_i(t)$ can be expressed as a linear combination of the other variables in the system plus a residual noise term:

$$X_i(t) = \sum_{j \neq i} B_{ij} X_j(t) + \varepsilon_i(t), \tag{22}$$

where B_{ij} denotes the direct influence of variable j on variable i, and $\varepsilon_i(t)$ is an independent noise term. Stacking all N equations, we obtain the simultaneous equations system

$$X(t) = BX(t) + \varepsilon(t), \tag{23}$$

where $B \in \mathbb{R}^{N \times N}$ is the coefficient matrix with zeros along the diagonal (no self-influence), and $\varepsilon(t) \sim \mathcal{N}(0, \Psi)$ is a vector of residual errors with covariance matrix Ψ. Rearranging,

$$(I - B)X(t) = \varepsilon(t). \tag{24}$$

The coefficient B_{ij} quantifies the instantaneous causal influence from X_j to X_i.

4.2 Acyclic and Cyclic Conditions in SEM

SEM defines a system of simultaneous dependencies among variables through the coefficient matrix B. To better understand the structural assumptions underlying this system—particularly the distinction between recursive (acyclic) and feedback (cyclic) interactions—we examine the graph induced by B.

Each variable X_i is represented as a node, and we draw a directed edge $j \to i$ whenever $B_{ij} \neq 0$, meaning that X_j directly influences X_i. In this way, B defines a directed graph $G(B)$ that encodes the dependency structure of the SEM.

A *cycle* occurs when a sequence of directed edges starts and ends at the same node, e.g.,

$$i_1 \to i_2 \to \cdots \to i_k \to i_1.$$

If no such cycles exist, the graph is called *acyclic*. In the SEM, this corresponds to a recursive system in which variables can be arranged in an order so that each variable depends only on earlier ones and not on itself through a feedback loop. Such an arrangement is called a *topological ordering*.

When the graph is acyclic, such an ordering always exists. This means that we can relabel the variables so that all directed influences go in one direction, for example,

$$X_1 \to X_2,\ X_2 \to X_3,\ \cdots,$$

with no backward dependencies. This relabeling can be represented by a permutation matrix P, which simply rearranges the variables. Under this transformation, the coefficient matrix becomes

$$\tilde{B} = PBP^\top, \tag{25}$$

where P corresponds to the chosen topological ordering. By definition of this ordering, if there is a directed edge $j \to i$ (i.e., $B_{ij} \neq 0$), then node j must appear before node i in the ordering, which implies $j < i$. Therefore, in the reordered matrix $\tilde{B}$, a nonzero entry $\tilde{B}_{ij}$ can only occur when $j < i$, so that

$$\tilde{B}_{ij} = 0 \quad \text{whenever } j \geq i,$$

and hence $\tilde{B}$ is strictly lower triangular. This triangular structure reflects the absence of feedback and provides a sequential interpretation of the SEM.

In contrast, if the directed graph $G(B)$ contains cycles, then no such permutation exists and the system exhibits *feedback* or *cyclic dependencies*.

4.3 Parameter Estimation

Given data $X = [X(1), X(2), \ldots, X(T)] \in \mathbb{R}^{N \times T}$ across T time points, the SEM coefficients are typically estimated by ordinary least squares (OLS) regression performed for each variable X_i against the others:

$$\widehat{B}_{i,-i} = \arg\min_{\beta \in \mathbb{R}^{N-1}} \|X_i - \beta^\top X_{-i}\|^2, \tag{26}$$

where $X_i = [X_i(1), \ldots, X_i(T)]^\top$ is the time series of the i-th variable and X_{-i} is the $(T \times (N-1))$ matrix of all other variables. The residual vector $\widehat{\varepsilon}_i$ is then computed as

$$\widehat{\varepsilon}_i = X_i - \widehat{B}_{i,-i}^\top X_{-i}.$$

Stacking across all i, we obtain the estimated coefficient matrix $\widehat{B}$ and residuals $\widehat{\varepsilon}$. This procedure can be written compactly as

$$\widehat{B} = \arg\min_B \|X - BX\|_F^2, \tag{27}$$

subject to $\mathrm{diag}(B) = 0$, where $\|\cdot\|_F$ is the Frobenius norm. The solution can be computed efficiently using pseudo-inverse operations and is robust to high-dimensional settings.

4.4 SEM-based Causality Testing

To test whether brain region X_j has a causal effect on X_i, we examine the null hypothesis

$$H_0 : B_{ij} = 0, \tag{28}$$

$$H_A : B_{ij} \neq 0. \tag{29}$$

The test is performed using a t-test on the regression coefficient $\widehat{B}_{ij}$, where the standard error is obtained from the OLS residual variance and the covariance of the design matrix:

$$\mathrm{se}(\widehat{B}_{ij}) = \sqrt{\widehat{\sigma}^2 \left[(X_{-i}^\top X_{-i})^{-1}\right]_{jj}}.$$

The test statistic is then

$$t_{ij} = \frac{\widehat{B}_{ij}}{\mathrm{se}(\widehat{B}_{ij})},$$

with corresponding p-value

$$p_{ij} = 2\big(1 - F_{t,\,T-N}(|t_{ij}|)\big),$$

where $F_{t,\,T-N}$ is the cumulative distribution function of a t-distribution with $T-N$ degrees of freedom. Collecting these values across all pairs (i, j) yields a matrix of statistical significance that summarizes the strength of instantaneous effects in the SEM.

Analogous to the Granger causality matrix, we summarize the results in a matrix $S = (S_{ij})$ defined by

$$S_{ij} = p\text{-value of test for } B_{ij} = 0, \tag{30}$$

Table 1: Comparison of SEM and Granger Causality (GC).

Aspect	**Granger Causality**	**SEM**
Temporal vs. Instantaneous	Captures *lagged* influences; past values improve prediction.	Quantifies *instantaneous* interactions at the same time point.
Models	Vector Autoregressive (VAR) models with lagged matrix A.	Coupled regressions with coefficient matrix B.
Inference	Tests $H_0 : A_{ij}^{(p)} = 0$; uses F-tests on full vs. restricted VAR models.	Tests $H_0 : B_{ij} = 0$; uses t-tests on regression coefficients.
Graph representation	GC matrix $G = (G_{ij})$ edges encode temporal causality.	SEM matrix $S = (S_{ij})$ edges encode instantaneous causality.

where each entry quantifies the statistical evidence for an instantaneous causal influence from X_j to X_i. Smaller values of S_{ij} indicate stronger evidence against the null hypothesis of no direct effect.

To facilitate interpretation and visualization, we convert this significance matrix into a weighted directed graph by defining

$$w_{ij} = 1 - S_{ij}.$$

Larger weights correspond to stronger contemporaneous causal relationships. The resulting weighted network provides a compact representation of instantaneous directional influences among variables, complementing Granger causality, which captures lagged dependencies.

4.5 Comparison Granger and SEM Causality

SEM and Granger causality provide complementary views of causality in multivariate systems. Table 1 highlights their main differences. Granger causality is inherently temporal, detecting whether one variable's past improves another's forecast, making it suitable for studying information flow across time. In contrast, SEM emphasizes contemporaneous dependencies, identifying direct instantaneous couplings between variables. This distinction is particularly important in neuroscience: SEM can highlight structural or effective connectivity at a single time point, while Granger causality uncovers dynamic propagation of activity across brain regions. Using both frameworks in tandem can thus provide a more complete characterization of causal interactions in complex systems.

Granger causality and SEM are primarily designed to estimate local, pairwise directional relationships through regression-based frameworks. They rely on assumptions such as linearity, sufficient sample size, and a well-conditioned covariance structure. As the number of variables increases,

the number of parameters grows rapidly, making estimation increasingly sensitive to noise, model specification, and multicollinearity. As a result, stable recovery of the underlying network structure becomes difficult in high-dimensional settings.

4.6 Limitation of SEM Causality

While the previous formulation provides a way to estimate instantaneous causal effects through the coefficient matrix B, inferring higher-order structures such as cycles from these estimates is substantially more challenging. In principle, if the true adjacency matrix were known exactly, one could apply standard graph-theoretic algorithms, such as Johnson's algorithm for enumerating all simple cycles (Johnson, 1975) or Tarjan's strongly connected components algorithm (Tarjan, 1972), to identify cyclic structures in the network.

However, in practice the adjacency matrix must be estimated from finite and noisy data, and this estimation step fundamentally limits the reliability of such approaches. In applications such as fMRI, where measurements are inherently noisy and temporally correlated, the estimated coefficients $\widehat{B}_{ij}$ obtained from SEM or Granger causality tend to be dense, with many small but nonzero values. These weak connections are often driven by noise or indirect effects rather than true causal interactions, making it difficult to distinguish meaningful structure from spurious associations.

This difficulty is further exacerbated in high-dimensional settings. When the number of variables is large relative to the sample size, the estimation problem becomes ill-conditioned due to multicollinearity and overparameterization. As a result, the inferred network structure is unstable, and small perturbations in the data can lead to large changes in the estimated adjacency matrix. In such regimes, *many competing models explain the data equally well, and the identification of cycles becomes ill-posed, since apparent feedback loops may arise from noise amplification rather than genuine system dynamics.*

Consequently, although graph-theoretic algorithms can enumerate cycles given a fixed network, their practical utility in SEM-based causality is limited by the difficulty of obtaining a reliable and interpretable estimate of the underlying adjacency structure.

5 Structural Causal Model

While SEM provides a system of regression equations that describe relationships among variables, it does not by itself specify how these relationships should be interpreted under external manipulation. In many scientific settings, the central question is not only whether variables are

associated (regardless of whether the association is directed), but what would happen if one variable were actively changed (e.g., through treatment or stimulation). Regression-based relationships alone cannot answer this question, since they are derived from observational data and may reflect confounding, indirect pathways, or equilibrium effects rather than direct causal influence.

To address this limitation, structural causal models (SCMs) extend SEM by explicitly encoding how variables are generated through structural assignments and how interventions modify these assignments (Pearl, 2009; Peters et al., 2017). In an SCM, each variable is expressed as a function of its direct causes and an exogenous noise term, thereby providing a generative description of the system rather than a purely associational one. This distinction is crucial: while SEM characterizes statistical dependencies, SCM introduces a formal mechanism for reasoning about interventions and counterfactual outcomes (Pearl, 2009).

An SCM consists of a set of structural equations of the form

$$X_i = f_i(\mathrm{Pa}(X_i), \varepsilon_i), \quad i = 1, \ldots, N, \tag{31}$$

where $\mathrm{Pa}(X_i)$ denotes the set of parent variables of X_i, f_i is a deterministic function, and ε_i is an exogenous noise variable. These structural assignments define the data-generating process and induce a directed graph (causal diagram) in which

$$X_j \to X_i \quad \text{whenever } X_j \in \mathrm{Pa}(X_i).$$

A key assumption is that these mechanisms are invariant across environments, which enables causal reasoning under interventions (Peters et al., 2016).

To obtain an SCM from SEM, each regression equation is reinterpreted as a *causal mechanism* that governs how a variable is generated from its parents. This reinterpretation allows one to move from associational relationships to a generative model of the system. Interventions are modeled using Pearl's do(·)-operator, which modifies the system by replacing the structural equation of a variable with a fixed value. For example,

$$X_k = c \quad \text{under } \mathrm{do}(X_k = c), \tag{32}$$

removes the dependence of X_k on its original parents and induces a new distribution over the remaining variables (Pearl, 2009). This provides a principled way to predict how changes propagate through the system.

In practice, the structural equations are typically unknown and must be inferred from data. This leads to two related but distinct tasks: *structure learning* (Spirtes et al., 2000), which aims to recover the parent sets $\mathrm{Pa}(X_i)$ and hence the causal graph, and *parameter estimation*, which quantifies the strength of causal effects. Structure learning methods include

constraint-based approaches based on conditional independence tests, score-based optimization over directed acyclic graphs, and functional approaches that exploit properties of noise distributions (Peters et al., 2017).

Once the structure is determined, causal effects are defined through interventions rather than associations. For example, the causal effect of X on Y is given by

$$\frac{\partial}{\partial x}\mathbb{E}[Y \mid \mathrm{do}(X = x)], \tag{33}$$

which may differ from regression coefficients unless strong assumptions such as absence of confounding hold. More generally, tools such as the backdoor criterion and do-calculus allow interventional distributions $\mathbb{P}(Y \mid \mathrm{do}(X = x))$ to be expressed in terms of observed data (Pearl, 2009).

Overall, SCM separates the representation of causal structure from the quantification of causal effects. The graph encodes the data-generating mechanisms, while causal effects are computed through interventions on these mechanisms. This separation distinguishes SCM from SEM, where structure and parameters are typically inferred simultaneously from observed associations.

5.1 SCM example

Consider three variables: X (treatment), Z (mediator), and Y (outcome). A linear SEM is given by

$$X = \varepsilon_X, \tag{34}$$
$$Z = aX + \varepsilon_Z, \tag{35}$$
$$Y = bZ + cX + \varepsilon_Y. \tag{36}$$

These equations can be written compactly in matrix form as

$$\begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ c & b & 0 \end{pmatrix} \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} + \begin{pmatrix} \varepsilon_X \\ \varepsilon_Z \\ \varepsilon_Y \end{pmatrix}. \tag{37}$$

To obtain an SCM, we reinterpret each equation as a structural assignment:

$$X := \varepsilon_X, \tag{38}$$
$$Z := aX + \varepsilon_Z, \tag{39}$$
$$Y := bZ + cX + \varepsilon_Y. \tag{40}$$

This defines a causal graph with edges $X \to Z$, $X \to Y$, and $Z \to Y$.

An intervention $\mathrm{do}(X = x_0)$ replaces the first equation by $X := x_0$, yielding

$$Z = ax_0 + \varepsilon_Z, \tag{41}$$
$$Y = bZ + cx_0 + \varepsilon_Y. \tag{42}$$

Substituting Z into Y gives

$$Y = (ab + c)x_0 + b\varepsilon_Z + \varepsilon_Y. \tag{43}$$

Thus, the total causal effect of X on Y is $ab + c$, decomposed into the indirect effect ab through Z and the direct effect c.

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